

Chapter 3

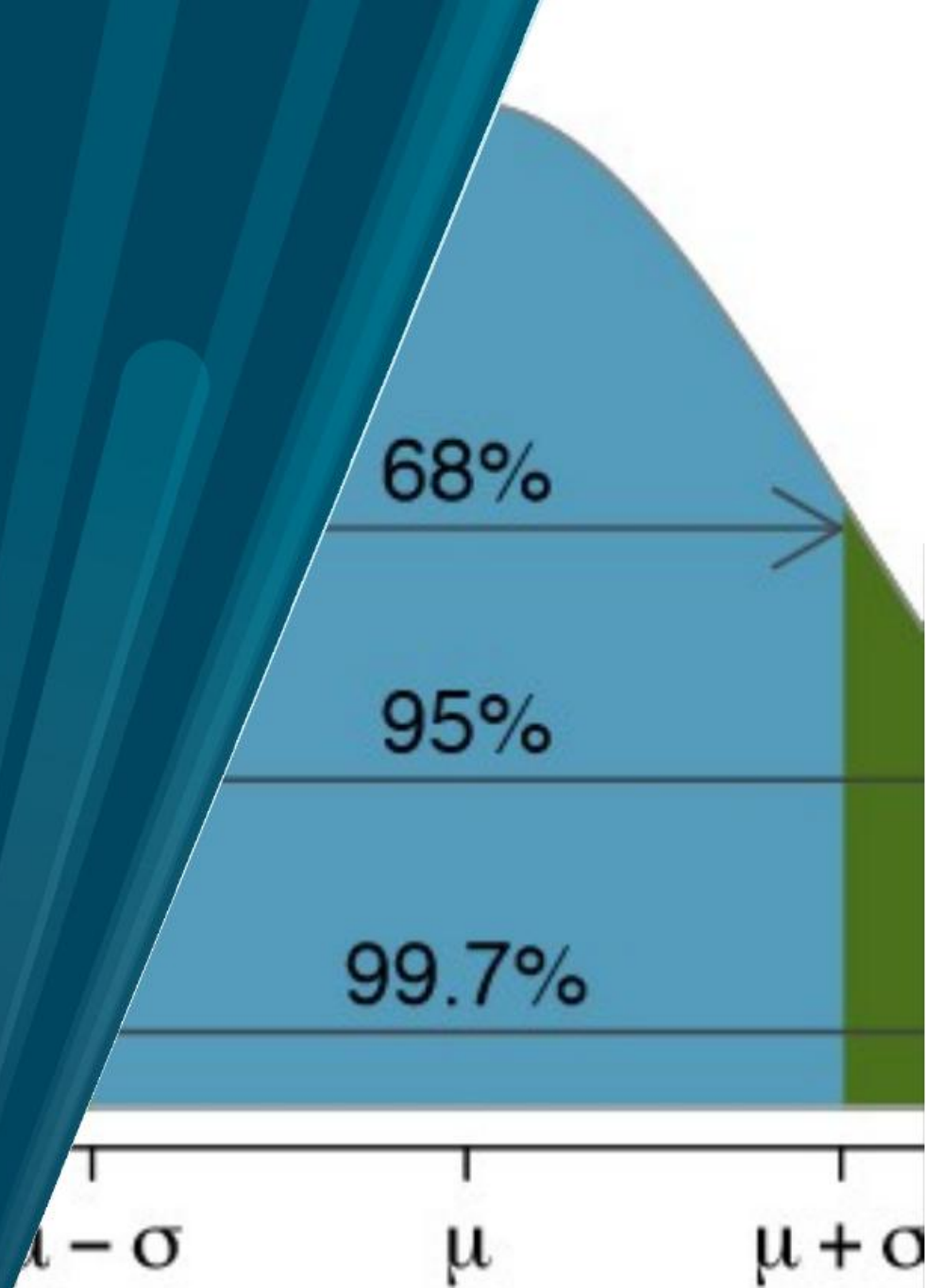
How certain are we?

Interval estimates and hypothesis tests

Sok-3081

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Results from *food* data

$$\hat{y}_i = 83.4 + 0.102 \cdot x_i$$

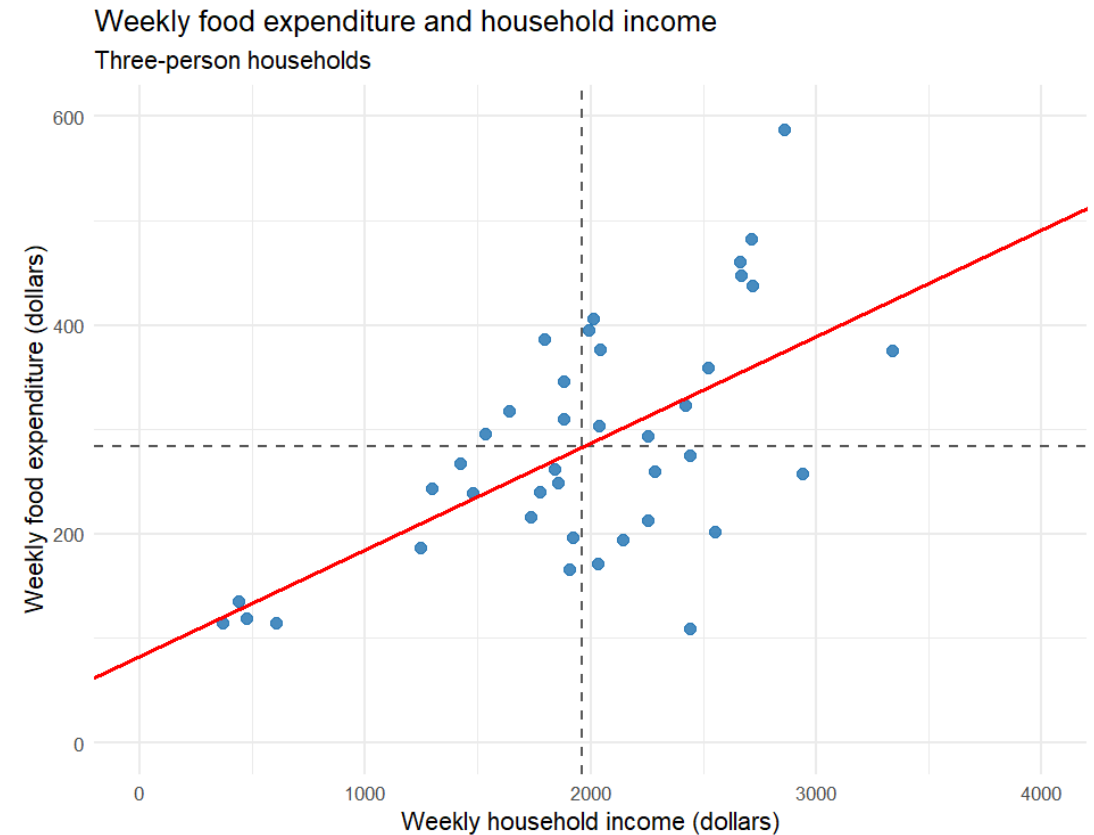
$$b_1 = \bar{y} - b_2 \cdot \bar{x} = 83.4$$

$$b_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = 0.102$$

$$SSE = \hat{\sigma}^2 = \frac{\sum_i \hat{e}_i^2}{n - 2} = 8013.2$$

$$\widehat{var}(b_1|x) = \hat{\sigma}^2 \cdot \frac{\sum_i (x_i)^2}{N \cdot \sum_i (x_i - \bar{x})^2}$$
$$\sqrt{\widehat{var}(b_1|x)} = se(b_1) = 43.4$$

$$\widehat{var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum_i (x_i - \bar{x})^2}$$
$$\sqrt{\widehat{var}(b_2|x)} = se(b_2) = 0.0209$$



The food model and the decision

$$\text{food_exp} = 83.42 + 0.1021 \text{ income}$$

USD 0.1021

Estimated increase in food
expenditure
per USD higher weekly income

USD 0.055

Minimum increase required
for the investment to be profitable

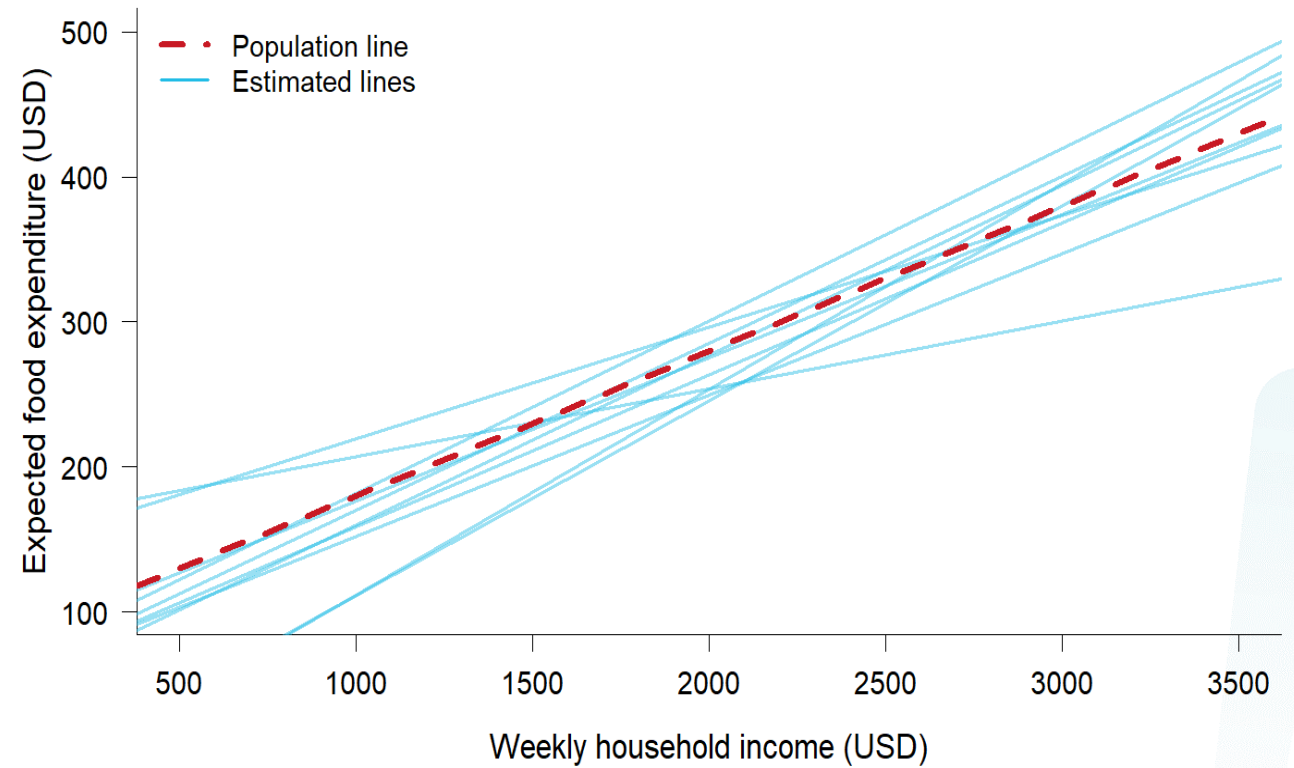
Should the supermarket invest?

Population parameter and sample estimate

$$\beta_2$$

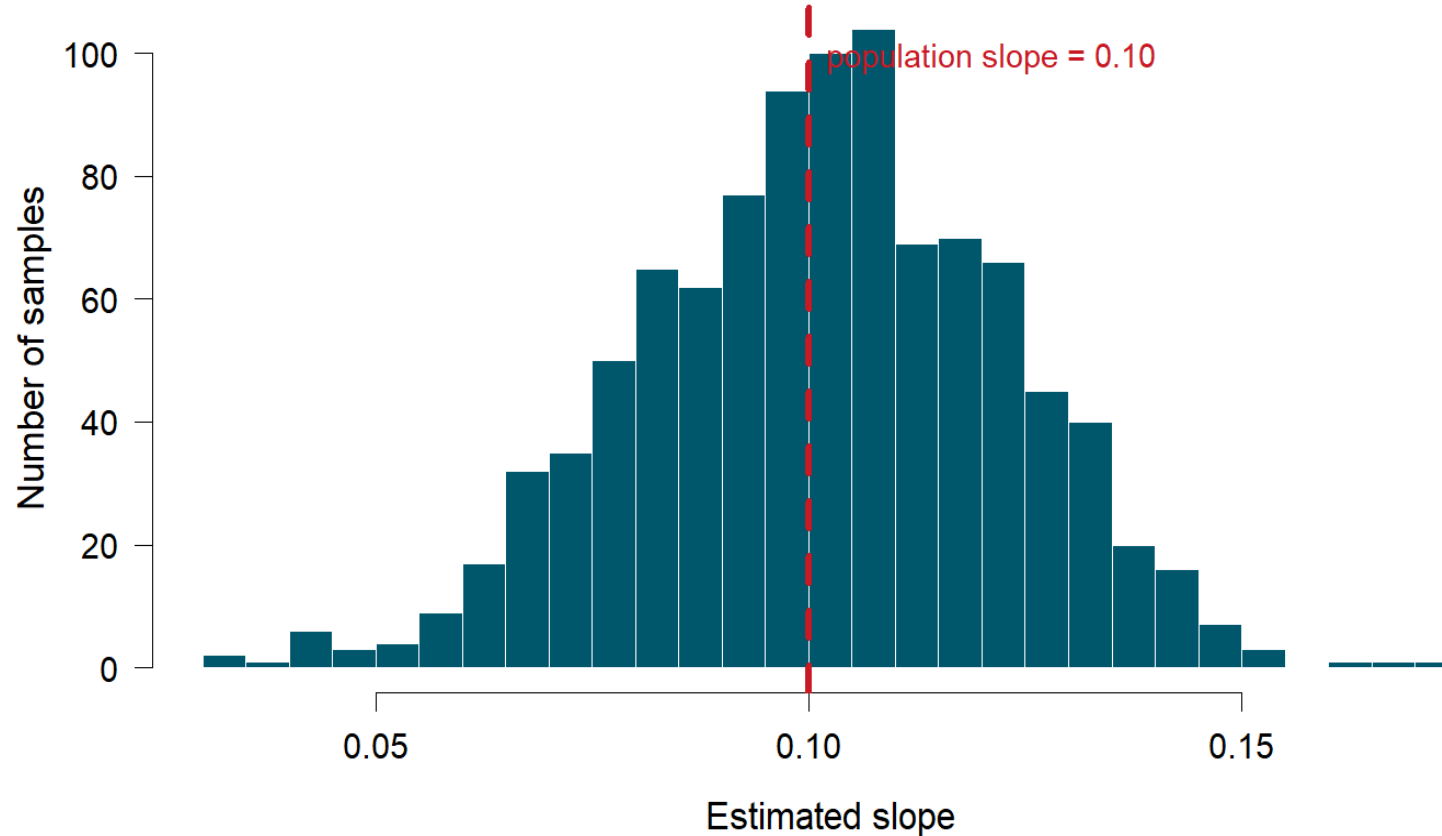
The fixed but unknown slope
in the population

One population value



**Each random sample produces a different
estimate b_2**

The sampling distribution of b_2

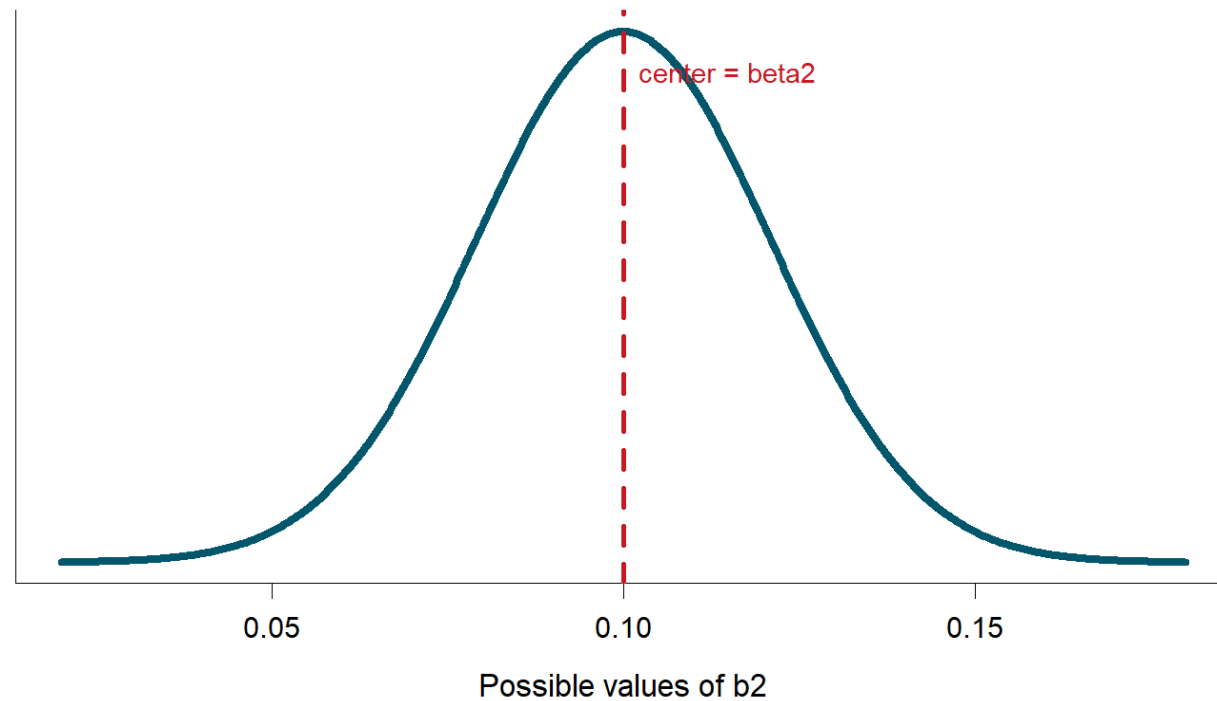


- Each random sample gives a new estimate b_2 .
- The estimates form a sampling distribution.
- The distribution is centered near the fixed parameter β_2 .

What changes if N increases from 40 to 200?

The distribution of the OLS slope estimator

$$b_2|x \sim N\left(\beta_2, \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}\right)$$



- Center: β_2
- Spread: smaller with less noise
- Spread: smaller with more variation in x
- Requires the simple regression assumptions, including normal errors

From a normal statistic to a t-statistic

If σ^2 were known

$$Z = \frac{b_2 - \beta_2}{sd(b_2)}$$

Z follows $N(0,1)$

In practice, σ^2 is unknown

Estimate σ^2 from the residuals

This adds uncertainty

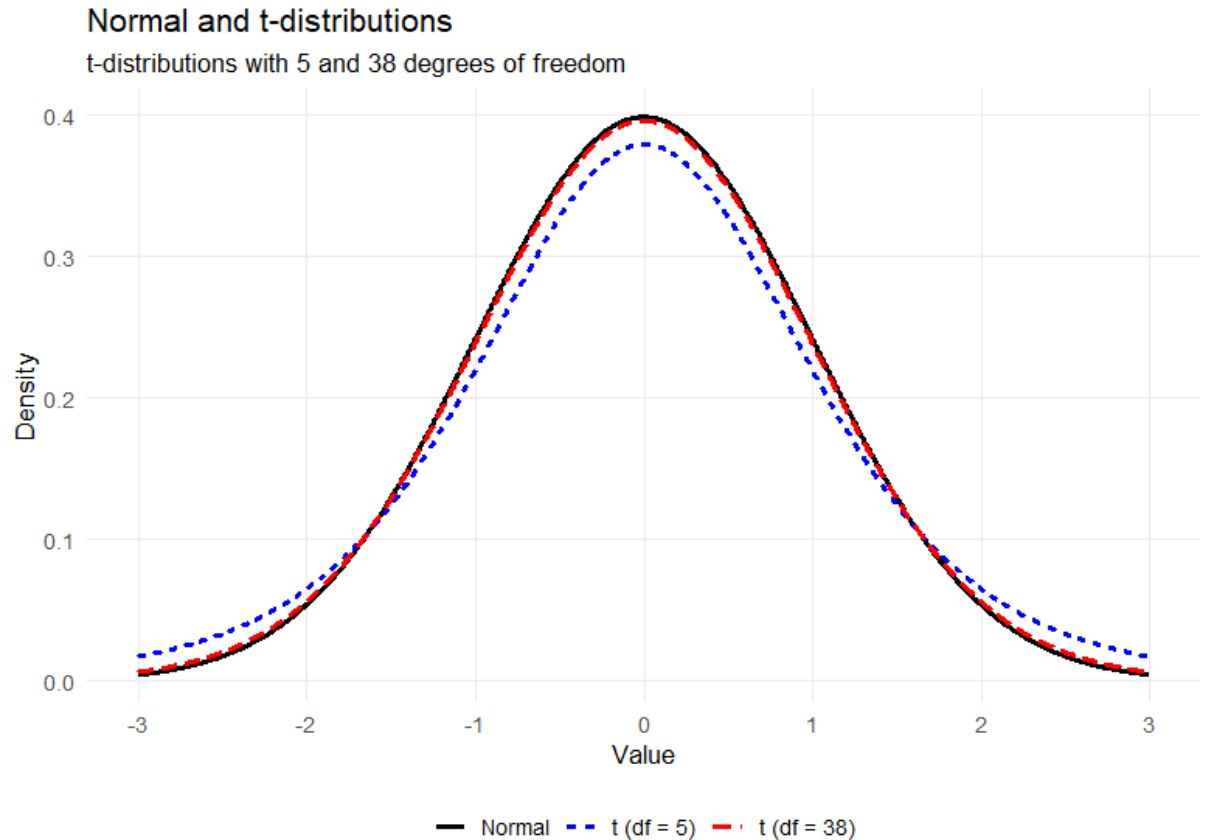
$$t = \frac{b_2 - \beta_2}{se(b_2)} \sim t(n - 2)$$

The t-distribution and degrees of freedom

- The t-distribution is symmetric around zero.
- Estimated σ gives thicker tails.
- The tails become thinner as the degrees of freedom increase.

Simple regression estimates two parameters

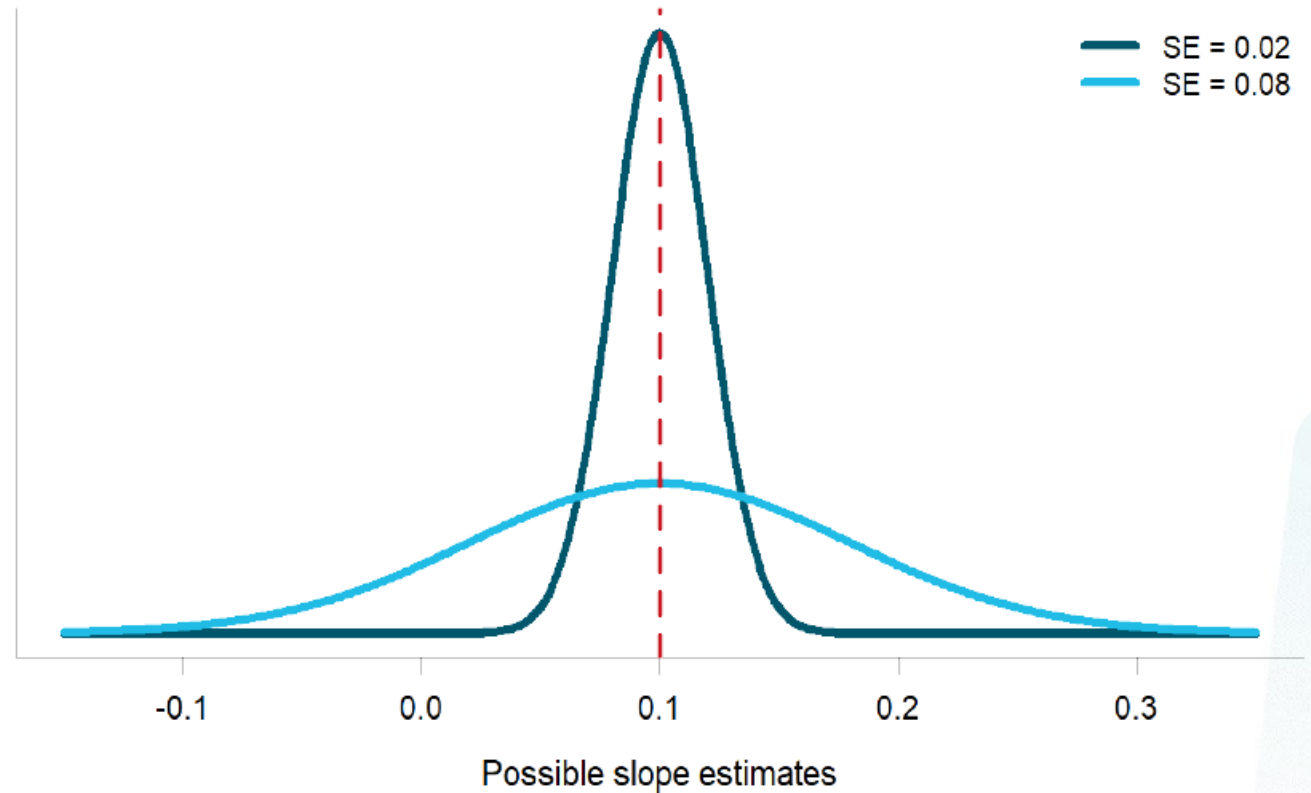
$N = 40$ gives $df = N - 2 = 38$



The standard error measures precision

se(b2)

Estimated standard deviation of the sampling distribution of b_2



Same estimate: $b_2 = 0.10$. Which study is more precise?

Estimates across samples

What changes when the sample grows from 40 to 200 households?

Predict before running R

- Mean of the estimates
- Spread of the estimates
- Average standard error

Run Parts A-C

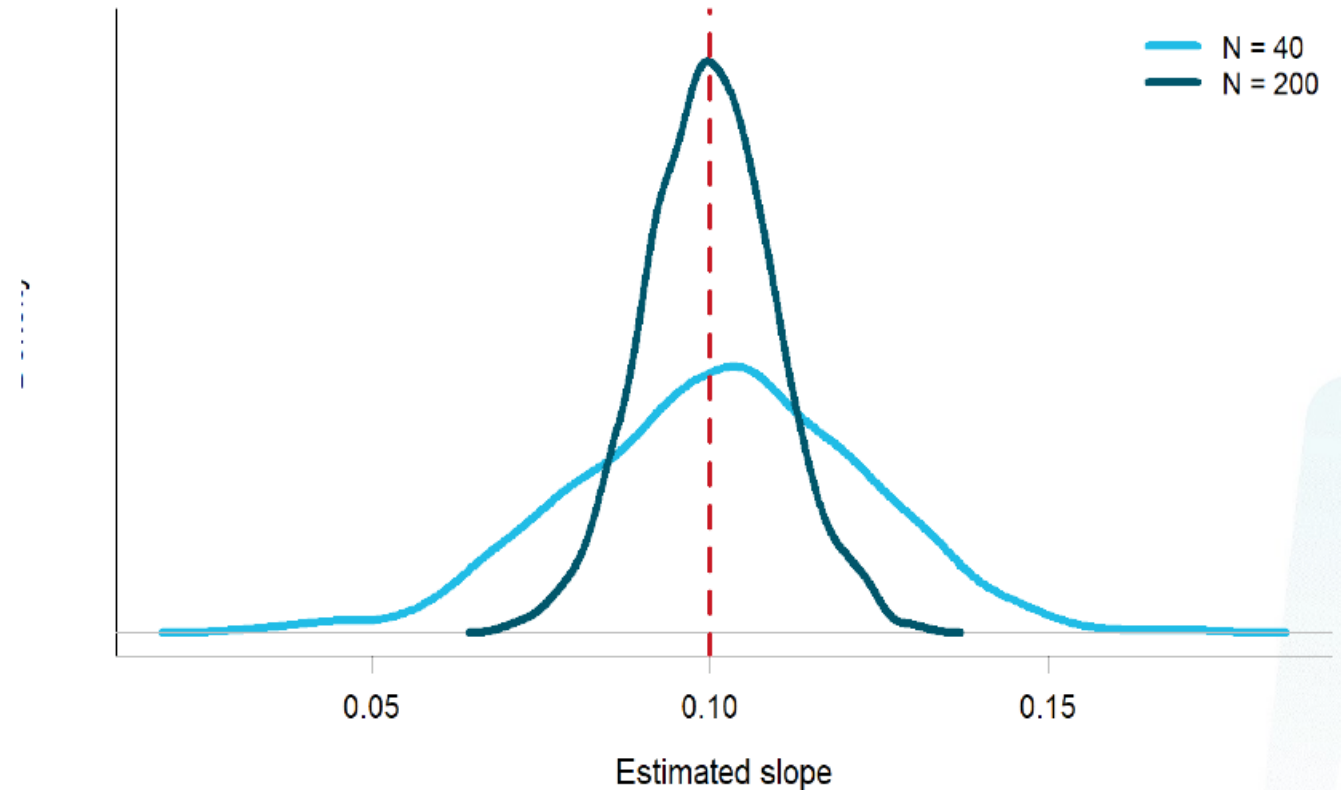
- Compare $N = 40$ and $N = 200$
- Explain the result in one sentence
- Use parameter, estimate, and standard error

A larger sample improves precision

Simulation results

N = 40
Mean b_2 : 0.1015
SD of b_2 : 0.0212
Mean SE: 0.0213

N = 200
Mean b_2 : 0.1001
SD of b_2 : 0.0097
Mean SE: 0.0094



Both distributions center near $\beta_2 = 0.10$. The larger sample is narrower.

What one sample can tell us

- β_2 is fixed and unknown.
- b_2 changes across samples.
- $\text{se}(b_2)$ estimates the sampling variability of b_2 .
- The t-distribution accounts for estimating σ .
- Larger samples usually give more precise estimates.

Next: Which values of β_2 are compatible with our sample?

From a t-statistic to an interval

$$t = \frac{b_2 - \beta_2}{se(b_2)}$$

For a 95% interval, choose critical values so that the central 95% of the t-distribution lies between $-t_c$ and t_c

$$P(-t_c \leq t \leq t_c) = 0.95$$

The next step is just algebra.

Goal: isolate the unknown parameter β_2 .

Derivation: construct the 95% interval

Start here:

$$-t_c \leq \frac{b_2 - \beta_2}{se(b_2)} \leq t_c$$

The interval estimator

$$b_2 \pm t_c \cdot se(b_2)$$

$$b_2$$

Center of the interval
Our point estimate

$$se(b_2)$$

Sampling uncertainty
Larger SE $\rightarrow$ wider interval

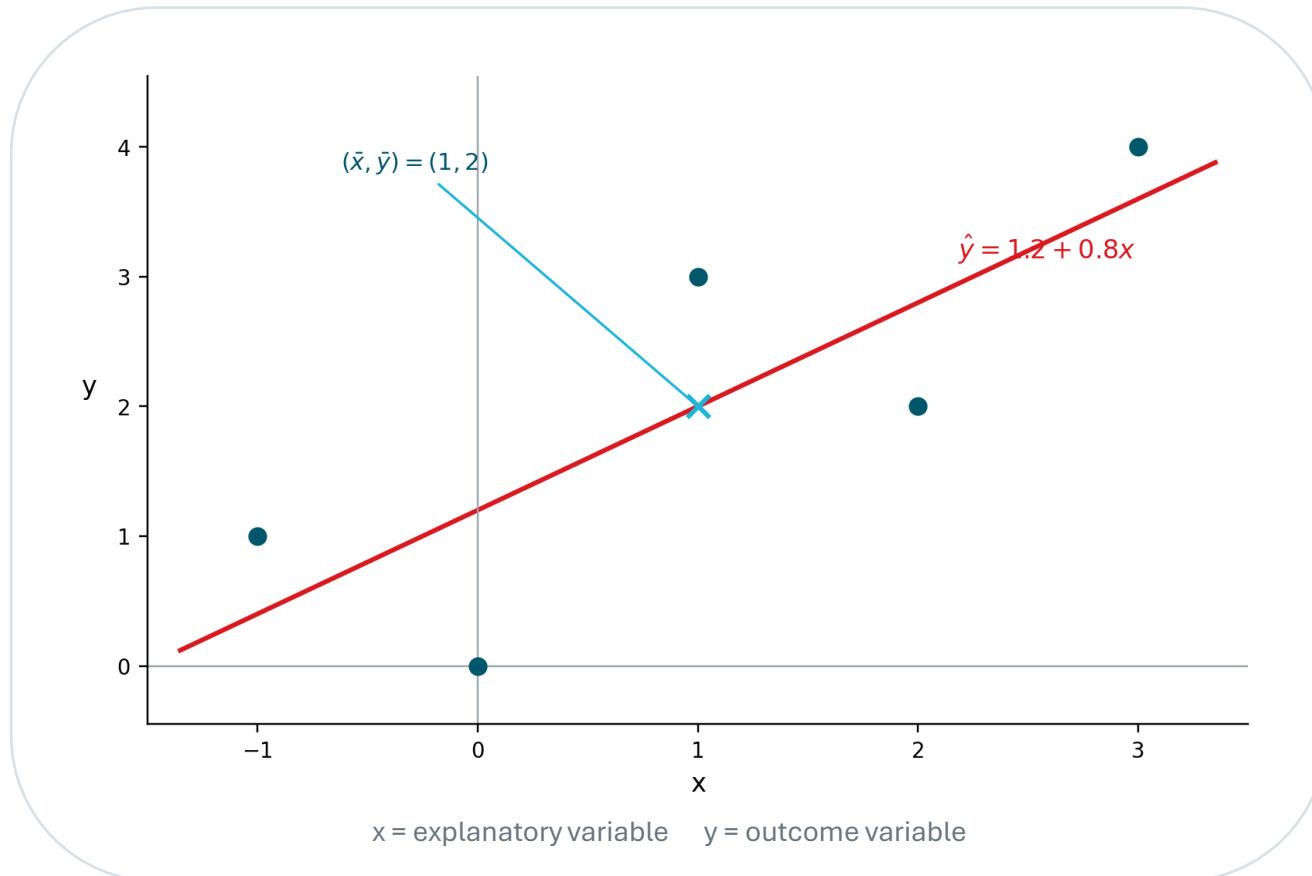
$$t_c$$

How much probability we want in the
procedure
Depends on confidence level and df

Estimate + uncertainty + chosen coverage procedure

We built OLS from these five observations

Everything below was calculated by hand before using `lm()`.



The data

x	y
-1	1
0	0
1	3
2	2
3	4

What we carry forward

Point estimate

$$b_2 = 0.80$$

Estimated sampling uncertainty

$$se(b_2) = 0.346$$

$N = 5 \rightarrow df = 3$

Fitted regression line

$$\hat{y} = 1.2 + 0.8x$$

Today: use b_2 and $se(b_2)$ to ask which values of the unknown population slope β_2 are compatible with this sample.

Back to our five-observation example

From Session 10, we already found:

$$b_2 = 0.80$$

$$se(b_2) = 0.346$$

$$N = 5 \rightarrow df = 3$$

Construct a 95% interval for β_2 .

- Find $t_{0.975,3}$.
- Insert it in $b_2 \pm t_c \cdot se(b_2)$.
- Interpret the width of the interval.

Use R as a calculator if you do not have a t-table:

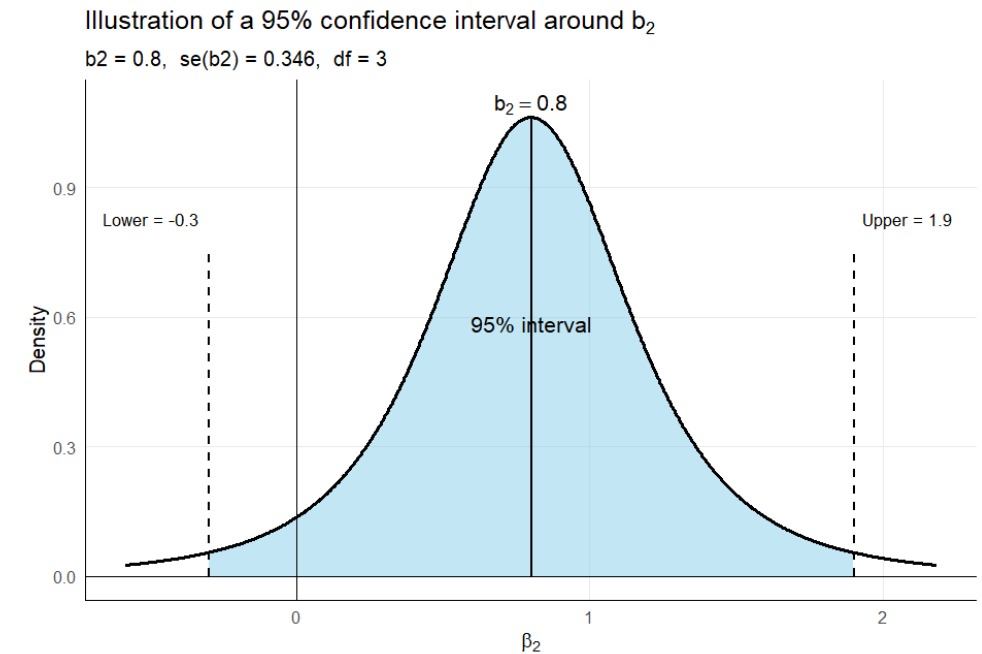
```
qt(0.975, df = 3)
```

Your calculation

$$b_2 = 0.8 \quad se(b_2) = 0.346 \quad df = 3 \quad t_{0.975,3} = 3.1824$$

95% interval: _____

What does the width tell us?



Manual first — then let R check it

```
# Critical value
t_c <- qt(0.975, df = 3)

# Manual interval
lower <- 0.80 - t_c * 0.346
upper <- 0.80 + t_c * 0.346
c(lower, upper)
```

```
# Same idea after estimating a model
model <- lm(y ~ x, data = small_data)
confint(model, "x", level = 0.95)
```

The function is convenient because we already understand the calculation behind it.

95% confidence is a property of the procedure

β_2

Fixed and unknown
There is one population value

Random sample

A new sample gives new b_2
and $se(b_2)$

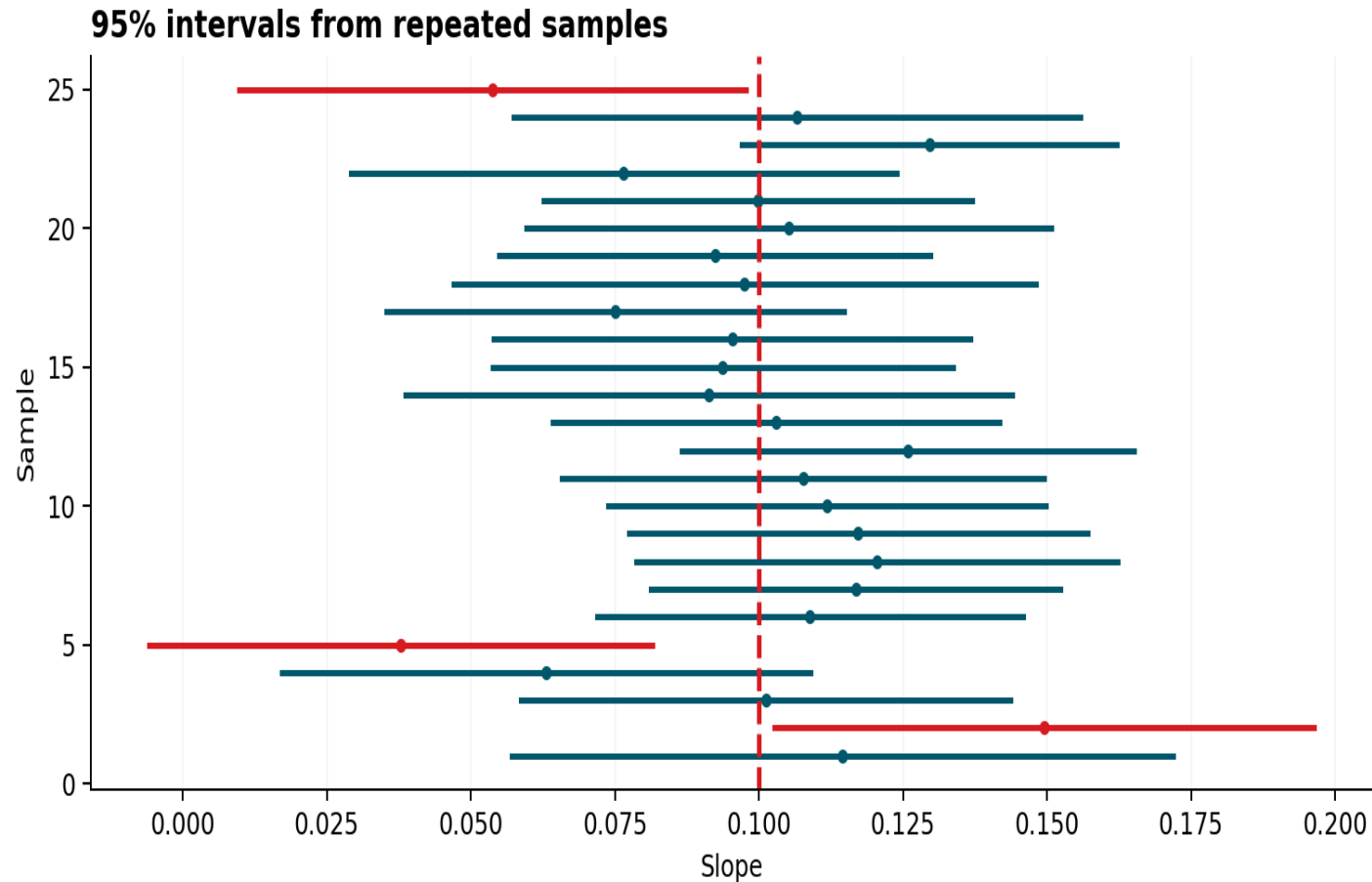
Interval

The endpoints change from
sample to sample

If we repeatedly sample and construct intervals in the same way, about 95% of the intervals contain β_2 .

The confidence is in the method — not in the particular interval after it has been calculated.

Repeated sampling makes the meaning visible



Dashed line

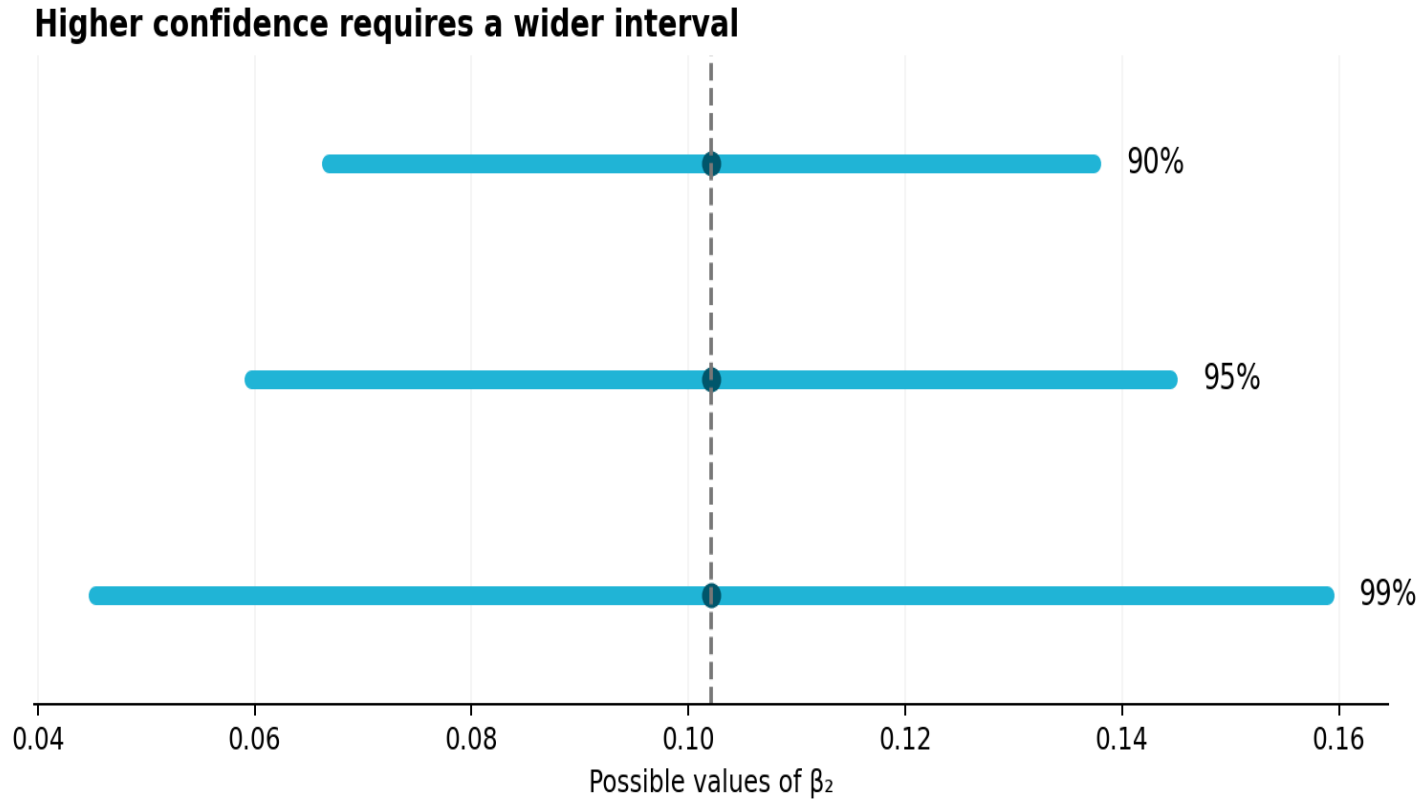
The fixed true β_2

Red intervals

Intervals that miss β_2

A particular interval either contains β_2 or it does not. The 95% statement describes the repeated procedure.

Higher confidence requires a wider interval



- Same estimate.
- Same standard error.
- Only the confidence level changes.

To capture the parameter more often, the procedure must allow a wider range of values.

Predict before running R: which interval is widest?

The investment threshold and the interval

Additional food expenditure per USD 100 higher income



Does the interval settle the investment decision?

New data: motel occupancy

Textbook computer exercise 3.19

$$MOTEL_PCT = \beta_1 + \beta_2 \cdot COMP_PCT + e$$

- Plot motel occupancy against competitor occupancy.
- Estimate the regression with `lm()`.
- Extract b_2 , $se(b_2)$, and df .
- Construct the 95% interval manually.
- Check with `confint()`.

Interpretation question

What does a one-percentage-point increase in competitor occupancy correspond to in predicted motel occupancy?

Work in pairs • 12–15 minutes

Motel debrief

1. Estimate

What is b_2 and what are its units?

2. Uncertainty

How large is $se(b_2)$?
How wide is the interval?

3. Interpretation

Which population slope values
are compatible with the sample?

Do not stop at “the interval is [lower, upper]”. State what the endpoints mean in the economic context.

Confidence is still about the procedure used to construct the interval.

Hypothesis testing

From a range of plausible values to one specific claim

Return to the supermarket decision

$$food_exp = 83.42 + 0.1021 \cdot income$$

Estimate

USD 0.102 more food expenditure
per USD higher weekly income

Decision threshold

Investment requires more than
USD 0.055 per USD

Question: Is the evidence strong enough to conclude that $\beta_2 > 0.055$?

From the economic question to H_0 and H_1

$$H_0: \beta_2 \leq 0.055$$

Not enough for investment

$$H_1: \beta_2 > 0.055$$

Large enough for investment

The alternative hypothesis determines the direction of the test.

For calculations, the boundary value 0.055 is used in the t-statistic.

The test statistic measures distance from the claim

$$t = \frac{b_2 - c}{se(b_2)}$$

Numerator

How far is the estimate from the hypothesized value?

Denominator

How much sampling variation is normal for b_2 ?

Interpretation: how many standard errors is the estimate from the value specified by H_0 ?

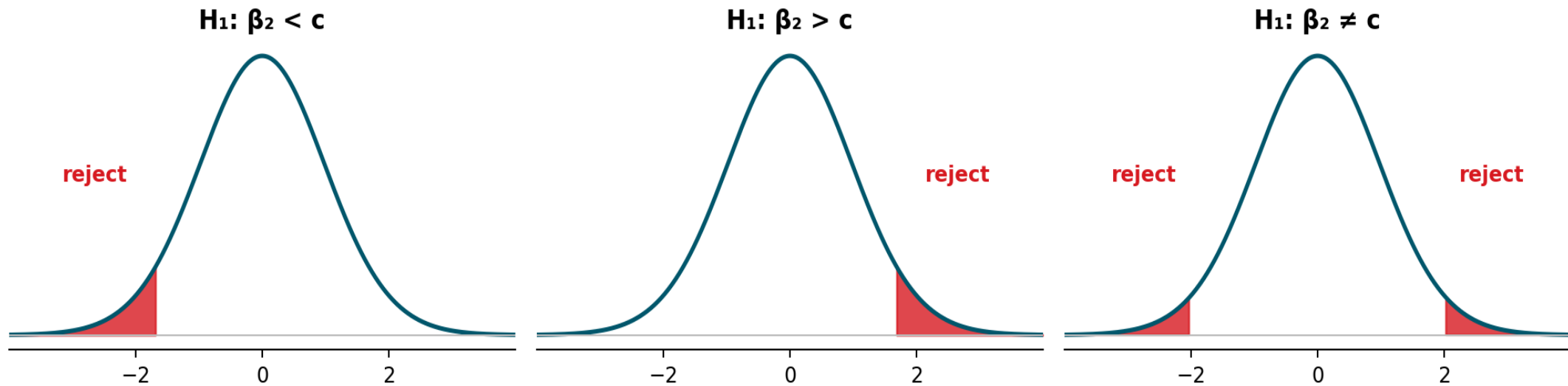
Calculate the supermarket t-statistic

$$b_2 = 0.1021 \quad c = 0.055 \quad se(b_2) = 0.0209$$

$$t = \frac{b_2 - c}{se(b_2)}$$

One idea, three alternatives

The alternative hypothesis determines the tail(s)



Same test statistic. What changes is where “unusual” values lie.

In this course we will mostly use two-sided tests, but the supermarket question is naturally right-sided.

A five-step routine

- 1 State H_0 and H_1
- 2 Calculate the test statistic
- 3 Choose α and the relevant tail(s)
- 4 Compare with a critical value or p-value
- 5 **State the conclusion in context**

The last step should use words from the economic question — not only H_0 , H_1 , and Greek letters.

Food example: same evidence, two thresholds

$$t = 2.25 \quad df = 38$$

$$\alpha = 0.05$$

Critical $t \approx 1.686$
 $2.25 > 1.686$

Reject H_0

$$\alpha = 0.01$$

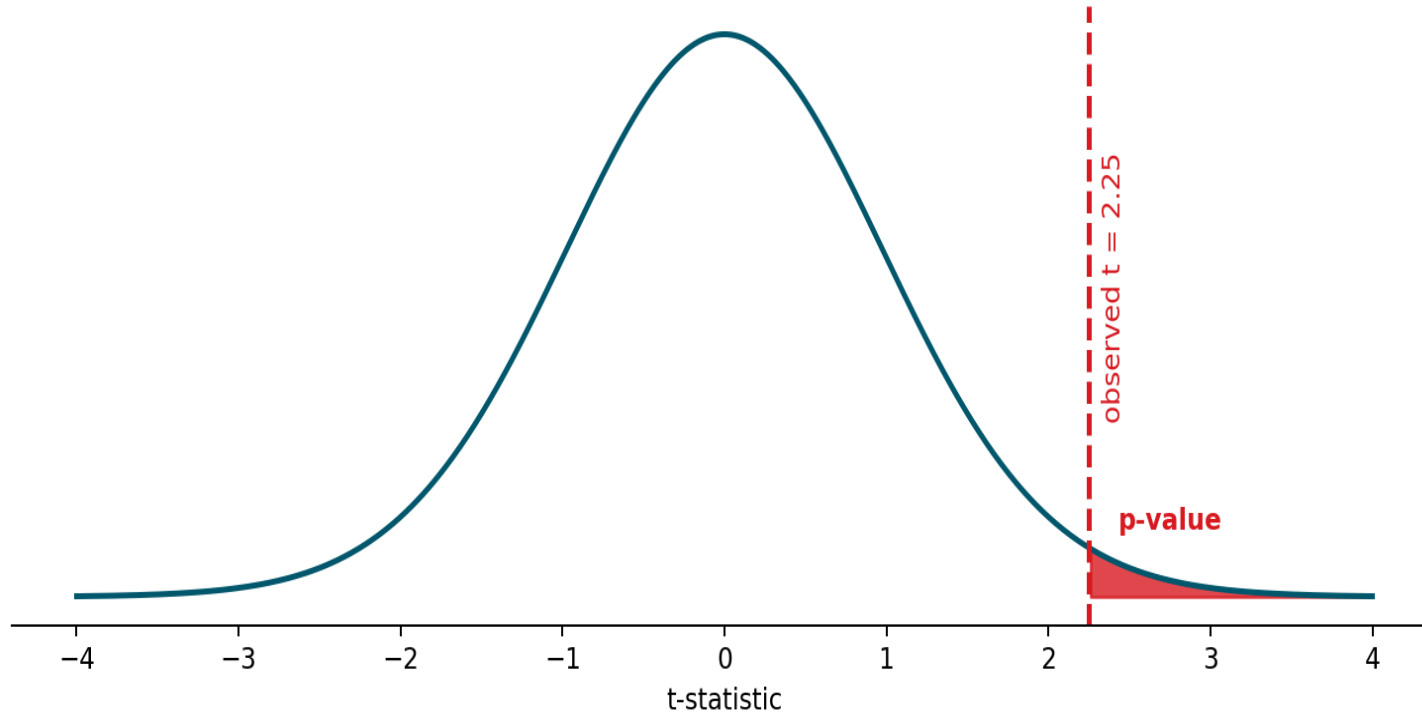
Critical $t \approx 2.429$
 $2.25 < 2.429$

Fail to reject H_0

The data did not change. The standard for how unusual the evidence must be changed.

The p-value asks how unusual our result is under H_0

Right-tail p-value: probability beyond the observed t under H_0



$$p = P(t \geq 2.25 | H_0)$$

Small p-value $\rightarrow$ the observed t is difficult to reconcile with H_0 .

Do not say: "the p-value is the probability that H_0 is true."

The p-value is the probability of obtaining a t-value as extreme or more extreme than the observed t-value, assuming that the null hypothesis is true.

R can calculate both routes

```
# Test statistic  
t <- (b_2 - 0.055) / se_b2  
  
# Critical value: right-tail 5% test  
qt(0.95, df = 38)  
  
# Right-tail p-value  
1 - pt(t, df = 38)
```

Decision rules

$$t \geq t_c \rightarrow \text{reject } H_0$$

$$p \leq \alpha \rightarrow \text{reject } H_0$$

Two ways of expressing the same evidence.

Calculate the p-value

Supermarket test

$$H_0: \beta_2 \leq 0.055 \quad H_1: \beta_2 > 0.055$$

Observed test statistic: $t = 2.25$

```
# Right-tail p-value  
p_value <- 1 - pt(t_food, df = 38)  
p_value  
# 0.015
```

What does 0.015 mean?

If the null value $\beta_2 = 0.055$ were true, the probability of obtaining a t-value of 2.25 or larger is about 1.5%.

$\alpha = 0.05$

$p < \alpha \rightarrow \text{Reject } H_0$

$\alpha = 0.01$

$p > \alpha \rightarrow \text{Fail to reject } H_0$

Same data, different decision threshold.

Your turn: a new study

A study reports:

$$b_2 = 0.084$$

$$se(b_2) = 0.016$$

$$df = 48$$

Test at $\alpha = 0.05$:

$$H_0: \beta_2 \leq 0.055 \quad \text{vs.} \quad H_1: \beta_2 > 0.055$$

- Calculate t .
- Find the critical value or p -value.
- State the conclusion in one sentence.

Use R as a calculator.

Two-sided tests and confidence intervals tell a consistent story

$$H_0: \beta_2 = c \quad \text{vs.} \quad H_1: \beta_2 \neq c$$

If c is inside the 95% interval

Fail to reject H_0
at $\alpha = 0.05$

If c is outside the 95% interval

Reject H_0
at $\alpha = 0.05$

This equivalence is for a two-sided 5% test and a 95% interval for the same parameter.

Errors are possible because we make decisions from samples

Type I error

Reject H_0 even though H_0 is true.

For the supermarket: invest because we conclude $\beta_2 > 0.055$ when the true slope is not above the threshold.

Type II error

Fail to reject H_0 even though the alternative is true.

For the supermarket: do not establish the threshold even though β_2 really is above 0.055.

α controls the probability of a Type I error under the null.

We will not develop power calculations here.

Type I and Type II error illustrated

H_0 : The person is **not** pregnant

H_1 : The person is pregnant

H_0 : is true

H_0 : is **not** true

Reject H_0



Keep H_0



Statistical significance is not economic importance

Statistical question

Is the estimated difference large relative to its sampling uncertainty?

This is what the t-statistic and p-value address.

Economic question

Is the magnitude large enough to matter for the actual decision?

This requires context, units, costs, benefits, and thresholds.

A tiny effect can be statistically significant in a large sample. A large estimate can be too imprecise to be statistically significant.

Inference does not create causality

Which statement is defensible?

A

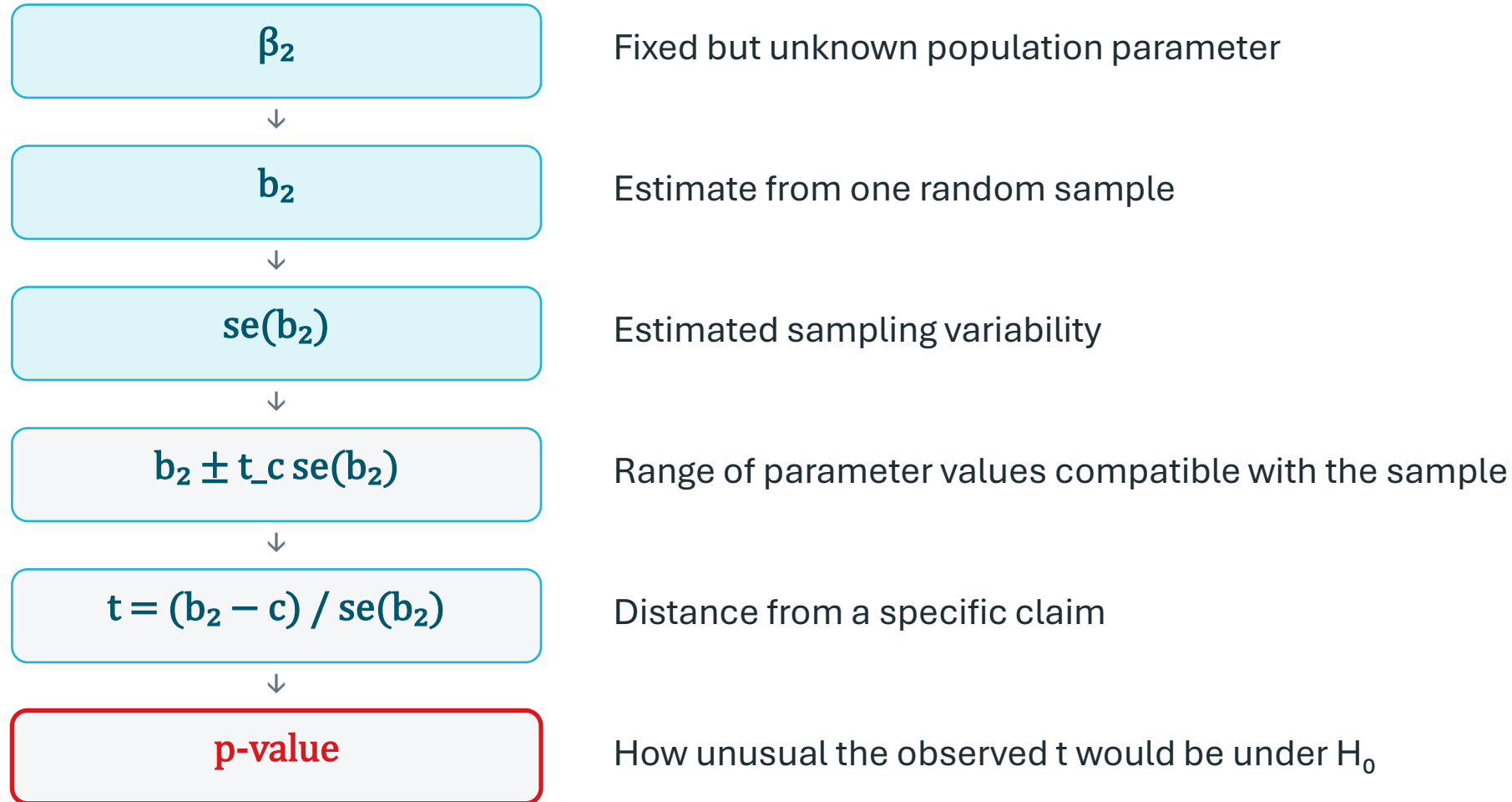
Raising a household's income by USD 100 causes food expenditure to rise by USD 10.21.

B

In this sample, higher income is associated with higher food expenditure; the estimated slope is 0.1021.

Confidence intervals and hypothesis tests quantify sampling uncertainty conditional on the model assumptions. They do not solve identification.

Chapter 3: one chain of reasoning



The same sampling uncertainty underlies both interval estimation and hypothesis testing.

Worked example — motel occupancy

Model

$$\text{MOTEL_PCT} = \beta_1 + \beta_2 \text{COMP_PCT} + e$$

What we estimate from the data

$$\text{MOTEL_PCT} = 21.340 + 0.865 \text{COMP_PCT}$$

SE

(0.203)

(12.907)

N = 25 df = 23

Question

Do the data provide strong evidence that the competitor's occupancy rate is positively associated with the motel's occupancy rate?

$$H_0: \beta_2 \leq 0$$

$$H_1: \beta_2 > 0$$

$$\alpha = 0.01$$



Right-tail

I will work this one through first.

Worked example — calculation and decision

1. Test statistic

```
motel_t <- (motel_b2 - 0) / motel_se_b2  
motel_t  
# 4.27
```

2. Critical value, right-tail $\alpha = 0.01$

```
qt(0.99, df = 23)  
# 2.50
```

3. p-value

```
1 - pt(motel_t, df = 23)  
# < 0.001
```

Decision

$$t = 4.27 > 2.50$$

and $p < 0.01$

Reject H_0

At the 1% level, the data provide evidence of a positive association between the two occupancy rates.

Always return to the economic question after the calculation.

Your turn — same data, a different claim

Test

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 \neq 1$$

$$\alpha = 0.01$$

What would $\beta_2 = 1$ mean economically?

Use the regression you already estimated.

```
motel_b2  
motel_se_b2  
motel_df
```

Tasks

- 1 Write down the test statistic.
- 2 Calculate t in R.
- 3 Find the two-sided critical value.
- 4 Calculate the two-sided p-value.
- 5 State the statistical and economic conclusion.

Do not use the p-value reported by `summary(lm())` — that tests $\beta_2 = 0$.

Your turn — a left-tail test with the motel data

A different regression

$$\text{MOTEL_PCT} = \delta_1 + \delta_2 \text{ REPAIR} + e$$

```
model_repair <- lm(motel_pct ~ repair, data = motel)
summary(model_repair)
```

Question: did the repairs reduce occupancy?

$$H_0: \delta_2 \geq 0 \quad H_1: \delta_2 < 0 \quad \alpha = 0.05$$

Tasks

- 1 Interpret the estimated REPAIR coefficient.
- 2 Calculate the t-statistic for the boundary value $\delta_2 = 0$.
- 3 Find the left-tail critical value.
- 4 Calculate the left-tail p-value.
- 5 State the conclusion in the context of the motel's claim.

Now the direction of H_1 points to the left.